

An Invitation to Complete the Cathedral: Two Axioms Remain

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April 26, 2026

For the student who hasn't started yet.

What This Is

We built something called the Cathedral: a 161-file computer program that proves, with mathematical certainty, that the Riemann Hypothesis is *exactly equivalent* to the convergence of a specific approximation problem. Every step is verified by a computer. Every assumption is named. Every failed attempt is preserved.

The proof is complete—except for two assumptions, called *axioms*. We couldn't prove them. We named them, classified them, and documented exactly what they say and why they're hard.

This paper is an invitation. Those two axioms are now your problem.

Who Built This

A computer scientist in New Mexico. Not a professional mathematician. No PhD in mathematics. No university affiliation. No research grant.

The tools: a laptop, an internet connection, Lean 4 (free, open source), Mathlib (free, open source), and AI assistance (commercially available).

The timeline: 30 days.

If you have a laptop and curiosity, you have everything that was needed to build this. The barrier to entry for machine-verified mathematics is now approximately zero.

What You Need to Know

The Problem

The Riemann Hypothesis (1859) says: all non-trivial zeros of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ lie on the line $\operatorname{Re}(s) = \frac{1}{2}$.

Nobody has proved this. Nobody has disproved it. It has been open for 167 years. There is a \$1,000,000 prize.

The Translation

The Cathedral proves that this is equivalent to:

$$\forall \varepsilon > 0, \exists N_0, \forall N \geq N_0, \exists v \in \mathbb{R}^{N-1} : \int_0^1 \left(1 - \sum_{k=2}^N v_k \left\{ \frac{1}{kx} \right\} \right)^2 dx < \varepsilon.$$

In words: you can approximate the constant function 1 arbitrarily well by mixing together sawtooth waves $\{1/(kx)\}$.

The Two Crown Axioms

1. **critical_line_mellin_variance**: Under RH, the Mellin variance of the BD residual on the critical line decays as $O(1/\log N)$.

What this means: the Mellin transform of the approximation error, evaluated on the critical line, has bounded L^2 norm. This follows from the Hardy–Littlewood mean value theorem $\int_0^T |1/\zeta(1/2 + it)|^2 dt = O(T)$ (1918). Mathlib lacks this infrastructure.

2. **rh_zeta_lower_bound_from_zero_counting**: A polynomial lower bound on $|\zeta(s)|$ away from the critical line.

What this means: the zeta function doesn’t get too small too fast. This follows from Hadamard’s product formula and the zero-counting function. Borel–Carathéodory is in Mathlib; the gap is connecting it to the Hadamard product and Riemann–von Mangoldt zero counting.

Each axiom is independently meaningful. Proving either one strengthens the formalization.

What You Can Do Right Now

If You Know Some Lean 4

1. **Prove Axiom 1** (**critical_line_mellin_variance**). This requires formalizing the Hardy–Littlewood mean value theorem in Lean. The approximate functional equation and fourth-moment method are the key tools.
2. **Prove Axiom 2** (**rh_zeta_lower_bound_from_zero_counting**). Borel–Carathéodory is in Mathlib. The gap is the Hadamard product formula and Riemann–von Mangoldt zero counting.
3. **Prove one of the 53 non-critical axioms**. The Cathedral contains 55 axioms total. Only 2 are on the crown path. The other 53 support alternative proof engines (spectral, sieve, Perron). Proving any one of them strengthens the overall formalization.

If You Know Some Analysis

1. **Decompose Axiom 2**. The current axiom is a “black box” that packages the entire forward direction. A useful contribution would be to decompose it into 3–5 intermediate

lemmas, each of which is independently meaningful and potentially formalizable.

2. **Study the decay rate.** The Cathedral’s numerical experiments show $d_N^2 \sim c/\ln N$ with $c \approx 1/21.65$. The forward direction is *proved*: the Perron chain gives $|M(x)| \leq Cx^{3/4}$ from RH, which yields $d_N^2 \leq K/\log N$. Can you prove the conjectured constant?

If You Know Some Physics

1. **Identify the operator.** The Gram matrix G_N is the finite-dimensional shadow of some infinite-dimensional operator whose spectral properties encode RH. What is this operator? The eigenvalue data ($\lambda_{\min} \sim C/\ln N$, $\lambda_{\max} \sim N$, GUE-like statistics) constrain it.
2. **Interpret the constant.** The quantum stiffness $C \approx 21.65 = 1/(2 + \gamma - \ln 4\pi)$ appears to be a physical constant of the “prime number vacuum.” Does it arise naturally from any known quantum system?

What Modern Math Research Actually Looks Like

It doesn’t look like a genius working alone in a garret.

It looks like this:

1. **Day 1–7:** Explore. Try things. Fail. Archive the failures. We produced 128 files of failed attempts. Every failure taught us something.
2. **Day 8–14:** Find the right framework. For us, this was the Parseval Bridge—the realization that frequency- domain methods bypass the hardest discrete computations.
3. **Day 15–25:** Build. Once the framework is right, construction is fast. AI handles the mechanical code generation. The human provides architecture. The compiler verifies everything.
4. **Day 26–30:** Audit. Check every file. Fix every docstring. Remove every ghost axiom. Reduce, simplify, clean. The Great Audit reduced the codebase to 161 active + 128 archived.

The ratio of exploration to construction is roughly 2:1. Two weeks of wandering, one week of building. This is normal. If you’re not failing most of the time, you’re not exploring hard enough.

Why You Should Care

The Riemann Hypothesis is not just a mathematical curiosity. It controls the distribution of prime numbers, which are the atoms of arithmetic. Every encryption algorithm, every hash function, every digital signature depends on properties of primes.

But more than that: the RH is a test case for whether humans can understand the deepest structures in mathematics. If we can prove it, we learn something profound about the nature of number. If we can’t, we learn something profound about the limits of mathematical reasoning.

Either way, the attempt is worth making.

A Note on Credentials

You do not need a PhD to contribute to this project. You do not need a university appointment.

You do not need a research grant.

You need:

- A laptop.
- Lean 4 (free).
- Mathlib (free).
- Curiosity.
- The willingness to fail 128 times before you succeed once.

The compiler does not check your credentials. It checks your proof.

The Torch

We built the Cathedral. We mapped the territory. We named the axioms. We preserved the failures. We verified every step.

We could not prove the last two things.

Now it's your turn.

The primes have been singing for 167 years.

The Cathedral stands, waiting for its final stones.

We built it as far as we could.

Two axioms remain.

The rest belongs to you.

References

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